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Using physics simulations to find targeting strategies in competitive tenpin bowling

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ABSTRACT

A new approach to finding the ideal location for a bowler to target on a bowling lane is demonstrated. To model bowling ball behavior, a system of six coupled differential equations is derived using Euler's equations for a rotating rigid body. The numerical solution to the equations of motion shows the path of the ball on the lane, demonstrates the phases of ball motion, and is ultimately used to output a plot that displays the optimal initial conditions for the shot trajectory that leads to a strike for a typical competitive bowler. When the bowler is modeled to be imperfect and some variance is included into the shot trajectory, it is shown that some targeting strategies lead to higher strike rates due to the “miss room” created from the inhomogeneity of the friction surface that results from the oil pattern.

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I. INTRODUCTION

Tenpin bowling remains as one of the most popular sports in the USA, with over 45×10^6 people participating regularly as of 2017.¹ With millions of dollars at stake every year in national competitions, significant research has been done to understand how players can achieve higher scores. Due to the complexity of the calculations and the vast number of variables that can affect the ball's trajectory, most of the research has focused on statistical analysis from empirical data instead of theoretical modeling. For example, the 2018 US Bowling Congress (USBC) Equipment Specifications Report used “37 bowlers with a full range of revolutions per minute (RPM) rates” rather than a computer model² and performed an analysis of ball tracking data from a professional bowling tournament.³

The literature on quantitative analysis of bowling physics is rare due to the many parameters involved but has been attempted by Fröhlich, Hopkins, and Huston over the past few decades.^{4–6}

Fröhlich and Huston created mathematical models that took into account the effects of a weight block within the bowling ball and provided simulation results for a small sample of parameter values, including effects of varying radius of gyration (RG), center of gravity (CG) offset from the geometric center of ball, and initial angular velocity. The simulations demonstrated the qualitative effects of changing certain variables but only assumed simple friction profiles that arise from the distribution of oil that is applied to the lane, known as an oil pattern.

Bowlers will typically look to play in an area of the lane such that they have a small region around their intended target that will result in their ball still hitting the headpin in the desired area at the desired angle so that the strike percentage remains high. This will allow for slight inaccuracies in their shot to not be punished greatly in terms of score. It has been shown that the optimal location for the ball to hit the headpin is ~6 cm offset from the center, and the optimal entry angle for the ball to be incident to the pin is ~6°.⁷

This paper aims to demonstrate target strategies that can be implemented by a bowler through a simulation that samples a large number of possible initial conditions, and it explores the effects of realistic friction profiles based on oil patterns that are used in current leagues and tournaments.^{8,9} The program can then provide the user with the best possible starting position, given their personal bowling numbers, such as ball speed, axis rotation, axis tilt, and angular velocity, by using experimental data to predict the outcome of the trajectories given by the mathematical model.

II. METHODS

A. Equations of motion

The equations of motion derived in this section describe rigid body rotation using a rotating frame of reference fixed to the ball. The derivation uses an approach through Euler's equations, where they are only dependent on the principal moments of inertia and not on the non-diagonal terms in the inertia tensor. This is important as the full inertia tensor of a reference frame fixed to space cannot be easily determined simply based on the published radius of gyration (RG) and differential values of each ball. Previous theories derived by Hopkins assumed a perfectly uniform spherical ball,⁵ while theories from Fröhlich required knowing the non-diagonal terms in the inertia tensor of the bowling ball.⁴ The CG offset, which is typically

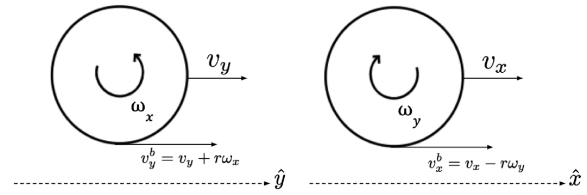


FIG. 1. Surface interactions between the ball and the lane along the $\hat{x}$ and $\hat{y}$ directions.

~1 mm away from the geometrical center of the ball, is assumed to be zero for this study in order to simplify calculations.

Ignoring air resistance, the only force due to the interaction between the ball and the lane is at their point of contact. The direction of the friction force on the ball is therefore purely dependent on the contact surface velocity $\vec{v}^b$, which has the following components:

$$v_x^b = v_x - r\omega_y, \quad (1)$$

and

$$v_y^b = v_y + r\omega_x, \quad (2)$$

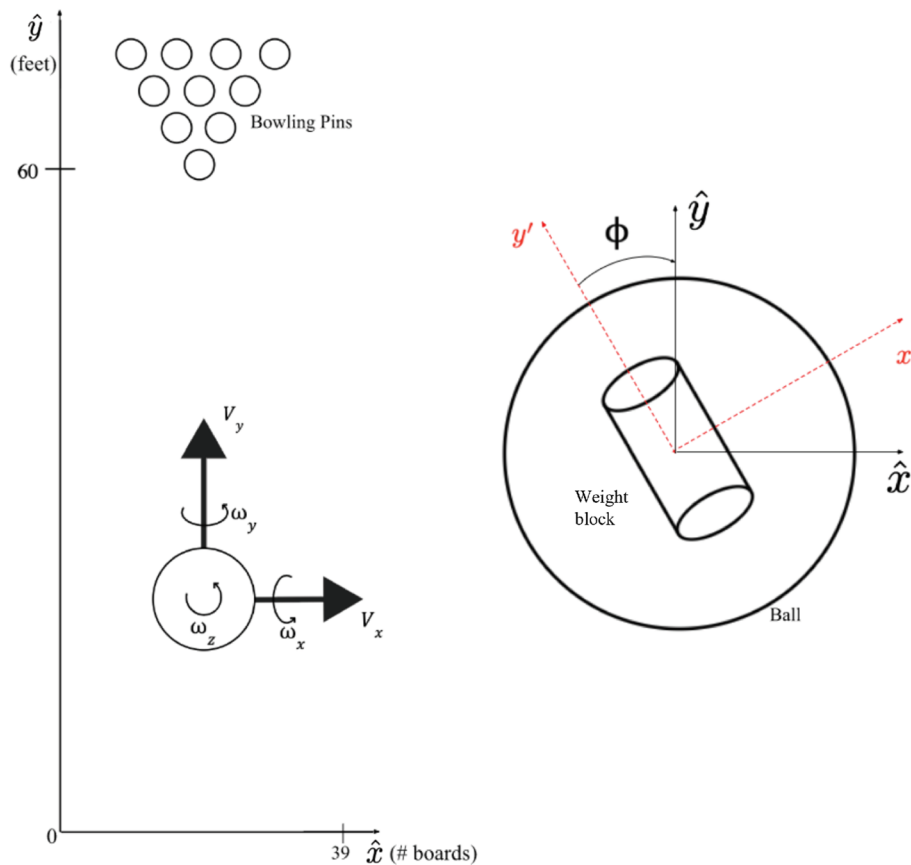


FIG. 2. Axis definitions with respect to the lane and the bowling ball. The $\hat{x}$ axis is measured in boards. A USBC approved bowling lane has 39 boards, each measuring ~2.73 cm or 1.07 in. The y' axis is aligned with the minimum moment of inertia axis of the weight block.

in the x and y directions, respectively, as illustrated in Fig. 1. The surface speed at the ball's point of contact with the lane can therefore be calculated using the formula given by

$$v^b = \sqrt{(v_x^b)^2 + (v_y^b)^2} = \sqrt{(v_x - r\omega_y)^2 + (v_y + r\omega_x)^2}. \quad (3)$$

Friction forces in the $\hat{x}$ and $\hat{y}$ directions are given by

$$F_x = m\dot{v}_x = -\mu mg \frac{v_x^b}{v^b}, \quad (4)$$

and

$$F_y = m\dot{v}_y = -\mu mg \frac{v_y^b}{v^b}, \quad (5)$$

respectively. The friction force then applies a torque to the ball given by

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{bmatrix} 0 \\ 0 \\ -r \end{bmatrix} \times \begin{bmatrix} m\dot{v}_x \\ m\dot{v}_y \\ 0 \end{bmatrix} = \begin{bmatrix} rm\dot{v}_y \\ -rm\dot{v}_x \\ 0 \end{bmatrix}. \quad (6)$$

Zero torque is applied in the z -direction as we assume that the ball contacts the surface at a single point. Combining these equations for torque with Euler's equations for a rigid body, in the frame oriented along the principal axes of the ball's weight block, we have

$$\tau_{x'} = I_{x'} \dot{\omega}_{x'} - (I_{y'} - I_{z'}) \omega_{y'} \omega_{z'}, \quad (7)$$

$$\tau_{y'} = I_{y'} \dot{\omega}_{y'} - (I_{z'} - I_{x'}) \omega_{z'} \omega_{x'}, \quad (8)$$

$$\tau_{z'} = I_{z'} \dot{\omega}_{z'} - (I_{x'} - I_{y'}) \omega_{x'} \omega_{y'}. \quad (9)$$

In order to align Eqs. (7)–(9) to the fixed axis of the lane so that they are in the same reference frame as Eqs. (4) and (5), they must be rotated through ϕ as principal moments of inertia are measured in the ball's reference frame (see Fig. 2). Then, we have a system of five first-order coupled differential equations for $v_x, v_y, \omega_x, \omega_y, \omega_z$ as

$$\dot{v}_x = -\mu g \frac{v_x - r\omega_y}{\sqrt{(v_x - r\omega_y)^2 + (v_y + r\omega_x)^2}}, \quad (10)$$

$$\dot{v}_y = -\mu g \frac{v_y + r\omega_x}{\sqrt{(v_x - r\omega_y)^2 + (v_y + r\omega_x)^2}}, \quad (11)$$

$$\begin{aligned} \dot{\omega}_x \cos \phi + \dot{\omega}_y \sin \phi = & \frac{1}{I_{x'}} \left[mr(\dot{v}_y \cos \phi + \dot{v}_x \sin \phi) \right. \\ & \left. + (I_{y'} - I_{z'})(-\omega_x \sin \phi + \omega_y \cos \phi) \omega_{z'} \right], \end{aligned} \quad (12)$$

TABLE I. List of parameters used to derive the ball trajectory.

Variable	Description
v_0, θ_0	Initial velocity and the launch angle of the ball. The launch angle θ_0 is measured relative to the y axis, typically ranging from 0° to 5°
$\omega_{x,0}, \omega_{y,0}, \omega_{z,0}$	Initial angular velocities imparted on the ball. A skilled bowler typically generates around 300–400 rpm, ⁴ while some bowlers using the two-handed style are able to generate ~600 rpm. The direction of rotation varies depending on bowling style and is typically determined using measurements of the “positive axis point”
$I_{x'}, I_{y'}, I_{z'}$	The principal moments of inertia, measured in a rotated frame so that the lowest inertia occurs along the y' axis. These values vary depending on the bowling ball and can be derived using three quantities given by the manufacturer, which are the radius of gyration (RG), differential (Diff.), and Intermediate Differential (Int. Diff.). $I_{y'} = m(\text{RG})^2$, $I_{x'} = m(\text{RG} + \text{Diff})^2$, and $I_{z'} = m(\text{RG} + \text{Diff} + \text{Int. Diff})^2$
μ	Kinetic friction coefficient between the ball's surface and the lane
ϕ	Angle between the y' axis and the fixed $\hat{y}$ axis of the lane (see Fig. 1)
x_0, y_0	Initial position of the ball. The x_0 coordinate is measured in boards, and the y_0 coordinate is measured in feet as shown in Fig. 2
m	Mass of the bowling ball. Most league and professional bowlers use masses between 6.3 and 7.3 kg (14 and 16 lbs)
r	Radius of the bowling ball. A typical ball has a radius of 10.85 cm

$$\begin{aligned} -\dot{\omega}_x \sin \phi + \dot{\omega}_y \cos \phi = & \frac{1}{I_{y'}} [mr(-\dot{v}_y \sin \phi - \dot{v}_x \cos \phi) \\ & + (I_{z'} - I_{x'}) (\omega_x \cos \phi + \omega_y \sin \phi) \omega_{z'}], \end{aligned} \quad (13)$$

$$\dot{\omega}_z = \frac{I_{x'} - I_{y'}}{I_{z'}} (\omega_x \cos \phi + \omega_y \sin \phi) (-\omega_x \sin \phi + \omega_y \cos \phi). \quad (14)$$

Since the weight block's orientation also changes as the ball's rotation changes, a sixth equation for ϕ given by

$$\dot{\phi} = \frac{d}{dt} \left(\arctan \frac{\omega_y}{\omega_x} \right) = \frac{\dot{\omega}_y \omega_x - \dot{\omega}_x \omega_y}{\omega_y^2 + \omega_x^2} \quad (15)$$

is needed to complete the system of equations. A description of the key parameters is given in Table I.

B. Oil patterns and strike chance

In competitive bowling, oil is applied to the lanes in patterns specifically designed to create challenging friction profiles that leave

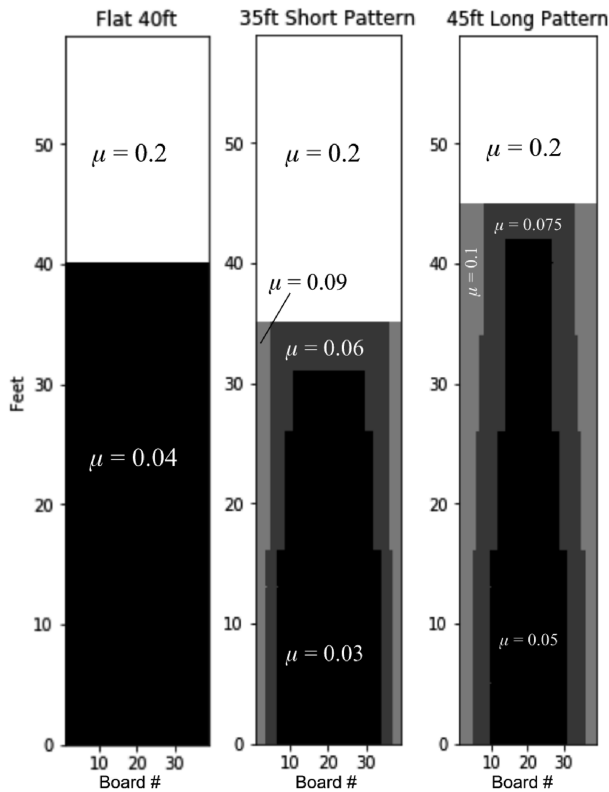


FIG. 3. Samples of typical competition oil patterns. Pattern lengths below 38 ft (11.6 m) are considered “short” and above 43 ft (13.1 m) are considered “long.” Friction coefficients for the flat pattern are estimates based on values given in Banerjee and McPhee.¹⁰ The 35 ft (10.7 m) and 45 ft (13.7 m) patterns are rough approximations of PBA Cheetah and Shark patterns, respectively, with their μ values based on the total volume of oil laid onto the lanes.

varying margins for error.^{8,9} Sample oil patterns are illustrated in Fig. 3. During competition, bowlers have to make educated guesses with shot features, such as ball choice, ball speed, and where to play on the lane, in order to maximize their chance of striking.

Two of the most significant factors that contribute to the chance of a strike are the entry position and entry angle. Entry position is the $\hat{x}$ coordinate of the ball’s position at the moment it makes impact with the pins ($\hat{y} \approx 60$ ft). Entry angle is the angle that the ball trajectory makes relative to the vertical at $\hat{y} \approx 60$ ft. The ideal position of entry is between 4 and 12 cm (1.57 and 4.72 in. or 1.5 and 4.4 boards) offset from the center of the lane, typically referred to by bowlers as the “pocket.” A 2009 empirical study conducted by the USBC provided the strike percentages at various entry angles and positions.⁷ These data are incorporated into the simulation, allowing it to calculate a score based on the ball’s trajectory.

C. Miss room for an imperfect bowler

In real life, no bowler can hit their target with 100% accuracy. The best professionals can get to within around 0.1° from their intended initial launch angle, which can correspond to a difference of a few centimeters further down lane. As shown in Fig. 4, this change will significantly affect the chances of a strike. In this study, the inaccuracy is modeled using a Gaussian distribution of starting angles, with the standard deviation equal to 0.1° .

III. RESULTS AND DISCUSSION

The equations of motion are solved subject to the following initial conditions that are typical for a competitive bowler. It is assumed that the bowler releases the ball at the foul line ($y_0 = 0$ m); the angular velocity that they impart to the ball at release is 416 rpm, that is rotating about an axis aligned 45° to the $\hat{x}$ axis and 13.3° to the $\hat{x} - \hat{y}$ plane, which gives $\omega_{x,0} = -30$ rad/s, $\omega_{y,0} = -30$ rad/s, and $\omega_{z,0} = 10$ rad/s; and the initial speed of the ball is $v_0 = 8$ m/s (17.9 mph or

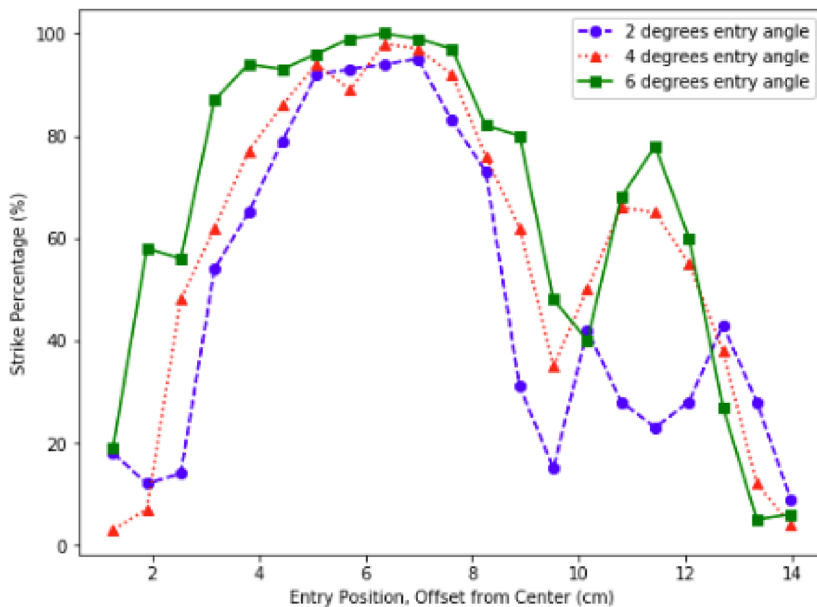


FIG. 4. Graph of strike percentages at 2° , 4° , and 6° entry angles as a function of entry position. Reproduced (with permission) from the 2009 USBC Bowl Expo presentation.⁷ Overall strike percentage is highest at 6° and lowest at 2° .

28.8 kph). The parameters x_0 and θ_0 are varied to find the different strike results. The properties of the bowling ball itself are as follows: RG = 6.35 cm, Diff. = 0.1 cm, Int. Diff. = 0 cm, and $m = 6.8$ kg.⁴

A. Phases of ball motion

The motion of the bowling ball on the lane can be categorized into the two classical phases of sliding and rolling. During the sliding phase, which typically occurs when the level of friction is low, the contact point between the ball and lane is not stationary in the lane's frame of reference ($v_b \neq 0$). Due to the low friction and large rotational inertia imparted by the bowler during the motion of releasing the ball, the ball spends most of its travel path in the sliding phase. Once the surface speed decreases to zero, pure rolling phase begins as the contact point becomes stationary relative to the lane. The ball travels in a straight line toward the pins as no more torque is applied.

Figure 5 demonstrates this through a plot of v^b over time on a flat oil pattern, where the distribution of oil is uniform throughout the area of the lane that is oiled.

B. Finding the ideal targeting strategy

In order to find the initial position x_0 and launch angle θ_0 that can lead to a strike, the outcomes for all possible x_0 values (0–39 boards) and starting angles θ_0 between 0° and 6° relative to the y axis were calculated. The results of this simulation are shown in Fig. 6 for the 40 ft flat oil pattern.

The simulation results can then be combined with the strike chance plots in Fig. 4, to determine the exact chances for these “pocket hits” in the black region to result in a strike. In order to incorporate the “imperfect bowler” idea from Sec. II C, a weighted average is calculated based on a Gaussian distribution of starting angles.

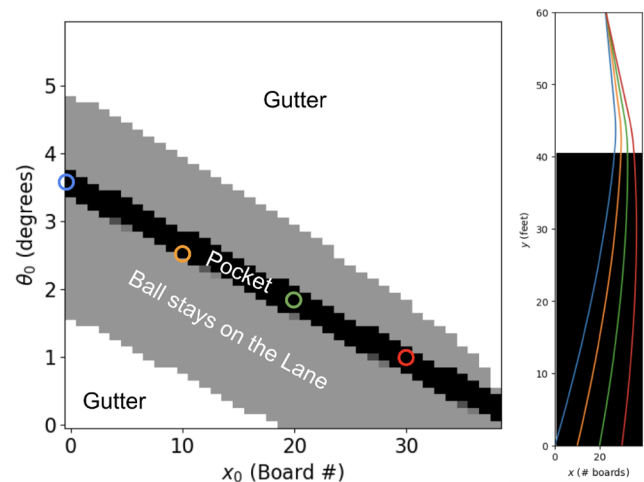


FIG. 6. 2D plot of simulation results, at all possible x_0 values and θ_0 between 0° and 6° relative to the y axis. The black pixels correspond to a pocket hit (4–12 cm off-center at entry). The gray areas correspond to when the ball does not go into the gutter but does not hit the pocket area. A sample set of paths that hit the pocket is shown in the right figure.

Figure 7 shows that there does not seem to be a strong preference toward any starting position for the flat pattern, with only a narrow region of possible starting conditions all exhibiting similar shades, which correspond to a similar probability of getting a strike. For the short pattern, however, a much wider area of viable starting positions can be found to have a high strike percentage near the turning point of the v-shaped probability curve, which arises from the inhomogeneous oil distribution near the edge of the lane. The

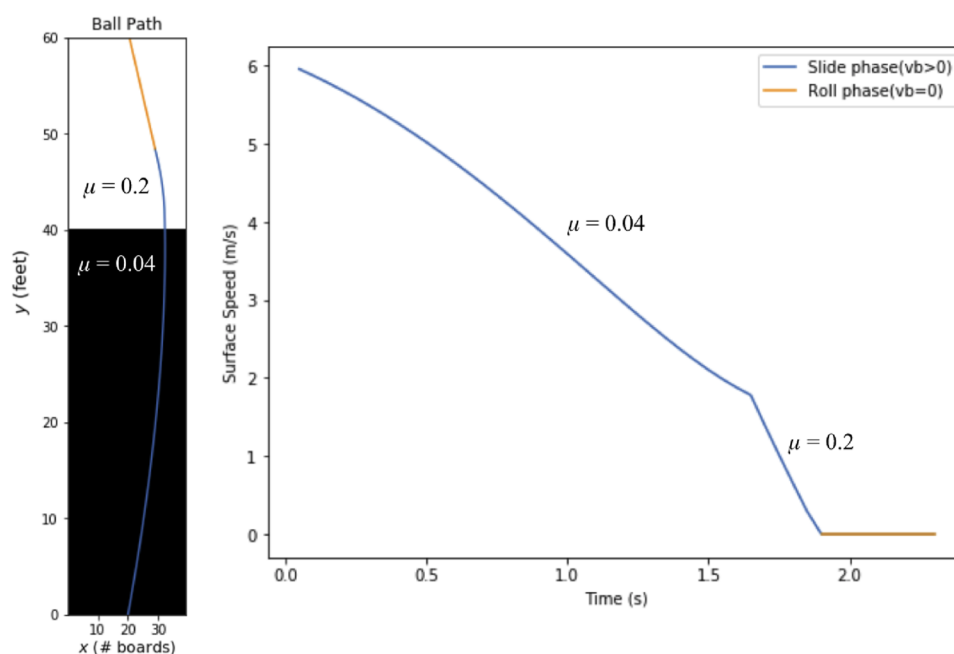


FIG. 5. Ball's surface speed over time and the corresponding locations in the ball path, starting at board 20 with an initial launch angle of 3° on the 40 ft flat oil pattern. The rate of change of surface speed is closely related to the friction coefficient.

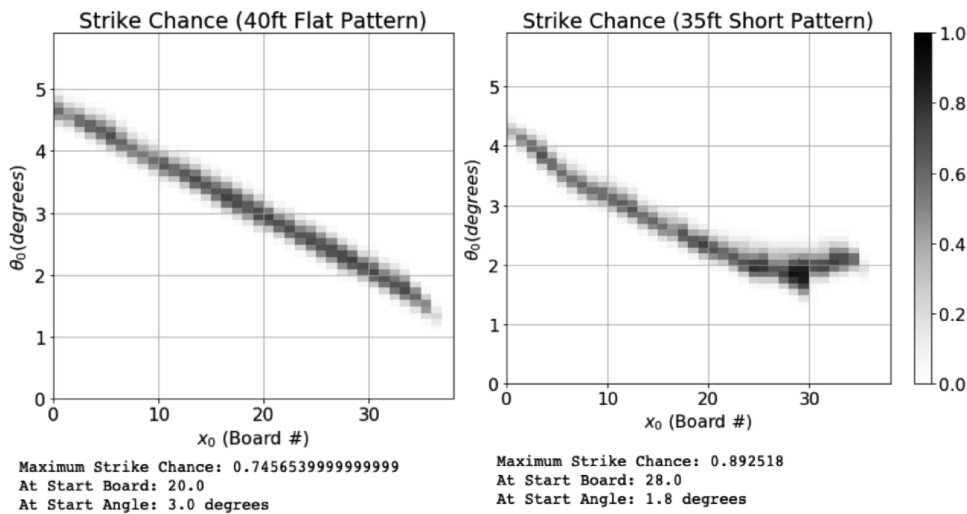


FIG. 7. Strike chance plots for the simulated bowler on flat (left) and short (right) patterns for different initial positions and launch angles of the ball given in boards and degrees, respectively (1 board ≈ 2.73 cm). The darker shades correspond to a greater chance of getting a strike.

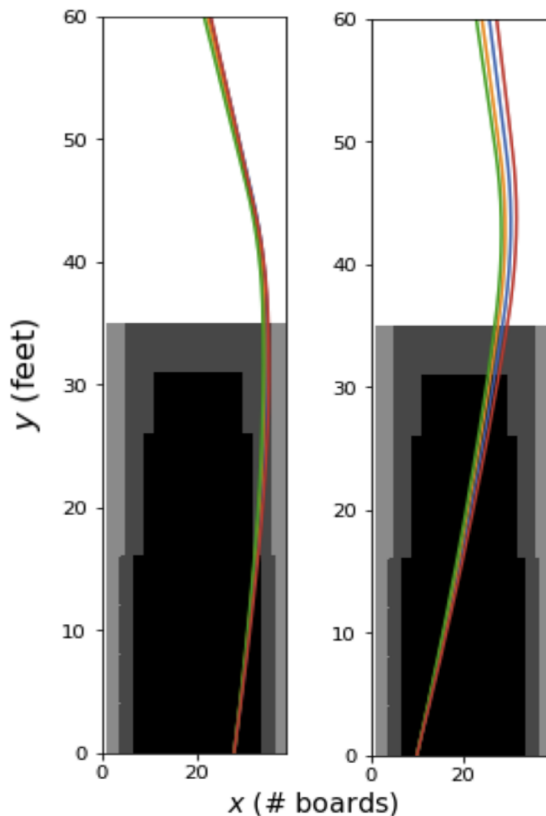


FIG. 8. Comparison between an “ideal line” ($x_0 = 28$ boards, $\theta_0 = 1.8^\circ$) and a “non-ideal line” ($x_0 = 10$ boards, $\theta_0 = 3.2^\circ$). The plotted paths represent small deviations ($\Delta\theta_0 = \pm 0.05^\circ, \pm 0.15^\circ$) to the original input.

highest strike chance is also found in this region, with an estimated value of 0.89, much larger than 0.75 found in the flat pattern. We define the “ideal line” on the short pattern to be the starting board and angle found by the simulation: $x_0 = 28.0$ boards; $\theta_0 = 1.8^\circ$.

The significance of the “maximum strike chance” is more clearly shown in Fig. 8. Even though both the “ideal line” ($x_0 = 28$ boards, $\theta_0 = 1.8^\circ$) and the “non-ideal line” ($x_0 = 10$ boards, $\theta_0 = 3.2^\circ$) lead to a pocket hit, small deviations in starting angle due to human error results in a much larger deviations in the final entry position for the “non-ideal line” (8 cm, compared to less than 3 cm for the ideal line).

The ideal line forms a path that closely aligns with the boundary between two friction zones of the short oil pattern. This makes sense qualitatively because if a bowler misses slightly to the right, the higher friction near the gutter would accelerate the ball to the left. Similarly, the lower friction in the center means a shot that misses slightly to the left will not hook early. In playing in this area, the bowler has some region around their intended target that they can hit while still getting the ball to finish in the desired region at the pins. While this result is not surprising to experienced bowlers, new bowlers often struggle to understand this concept and make adjustments that leave them with a much smaller margin for error.

IV. CONCLUSION

We have derived equations of motion that describe the evolution of a bowling ball with a fixed rigid internal weight block that travels on a lane with variable levels of friction that arise from the initial distribution of oil known as the oil pattern. For a given set of initial conditions that are typical of a competitive bowler, the equations of motion are solved for two different oil patterns, and an optimal strategy is formed that yields the largest target size while maintaining a high probability of getting a strike by matching the features of the ball path to experimental data. These results are easier to obtain than empirical studies and will be useful for both competitors and their coaches, as well as tournament oil pattern designers who would be able to gauge what the scoring pace of the tournament would be, given a group of bowlers.

In the theory derivations, several important factors that impact the ball’s motion could be considered for future study. For example, the model could be extended to capture the topography of the lane that could also influence optimal strategy. Such features

are neglected here as it is assumed that the lane is perfectly flat. The hardness of the bowling ball could also be captured by not assuming that the bowling ball contacts the lane at a point.

Further numerical investigation could also be performed to gain insight into the effect of features such as ball speed and axis rotation on optimizing the margin for error. Further experimental work is required to provide a more accurate set of friction coefficients that are specific to bowling ball surface, type of oil, and lane surface.

The model can be used to help bowlers choose the balls they would bring to the competition, as well as allowing oil pattern designers to determine whether they created a profile that is challenging but also fun to play in.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

S. S. M. Ji: Conceptualization (lead); Methodology (lead); Software (lead); Writing – original draft (lead). **S. Yang:** Conceptualization (equal); Formal analysis (equal); Project administration (lead); Writing – original draft (equal); Writing – review & editing (lead). **W. Dominguez:** Conceptualization (equal); Supervision (lead); Writing – original draft (equal); Writing – review & editing (supporting). **C. G. Hooper:** Conceptualization (equal); Formal analysis (equal); Project administration (lead); Writing – original draft (equal); Writing – review & editing (lead). **C. S. Bester:** Conceptualization (equal);

Supervision (lead); Writing – original draft (equal); Writing – review & editing (supporting).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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